The relationship between arbitrage in futures and spot markets and Bitcoin price movements: Evidence from the Bitcoin markets

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Abstract

We examine how investors arbitrage the Bitcoin spot and futures markets. Using intraday data of the Chicago Board Options Exchange (CBOE), we reconstruct the actual arbitrage condition that investors confront. We find that there are few arbitrage profit opportunities in "normal" markets, but large arbitrage profit opportunities arise during Bitcoin market "crashes."

JEL codes: G14, G12

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1. Introduction

In December 2017, the Chicago Board Options Exchange (CBOE) and the Chicago Mercantile Exchange (CME) launched Bitcoin futures contracts. While the efficiency of the Bitcoin market is widely discussed in the academic literature, the introduction of Bitcoin futures is expected to enhance the efficiency of the Bitcoin futures market by providing an additional instrument for investors. Although many studies reveal several types of inefficiency in the Bitcoin market, the arbitrage between the spot and futures market has not yet been examined. Testing the arbitrage between the spot and futures markets is the focus of several classical empirical studies and, to the best of our knowledge, we are the first to apply this to the context of the Bitcoin market. Our paper provides the evidence that supports the efficiency among the spot and futures market, which is consistent with the other futures market.

Using intraday data from CBOE, we find fewer arbitrage opportunities during normal times, that is, periods without crisis defined by more than 40% drop within one trading month. Although our finding is about a time series effect instead of a cross section, this result is consistent with Makarov and Schoar (2019a, b), who find price deviations in the cryptocurrency markets to be much smaller within than across countries while we show a few price deviations among the Bitcoin spot and futures market, which also belong to the same jurisdiction. However, we also confirm that market crashes may have led to arbitrage opportunities, especially right after the launch of Bitcoin futures. Our methodological approach follows the method established during the 1980s (e.g., Cornell and Reinganum, 1981), which evaluates if the deviation of the futures and forward price is less than the cost related to the arbitrage, such as the bid-ask spread. For robustness, we employ the futures data from CME, which provides daily but

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relatively longer data compared with the CBOE. In doing so, we find even fewer arbitrage opportunities.

Our study uses unique data. We use intraday data to consider the arbitrage condition. CBOE Bitcoin futures are cash-settled, referring to the auction price for Bitcoin in US dollars at 4 PM ET. Thus, we use the 4 PM ET price of Bitcoin futures to match the timing. Due to the high intraday volatility of Bitcoin futures, matching the timing with the intraday data is especially critical to test the arbitrage condition.

It must be emphasized that the CME introduced its futures one week after the CBOE bitcoin futures contract, and the CME bitcoin futures contract overtook the CBOE bitcoin futures contract in size-adjusted futures trading volume by March 2018. In fact, the CBOE has stopped offering new futures since June 19, 2019; therefore, the CBOE ceased Bitcoin futures trading. However, we use the data of CBOE in terms of arbitrage profit opportunities. The reason for focusing on the CBOE bitcoin futures contract in our study is to facilitate testing of arbitrage profits in a more ideal setup for the timing of the arbitrage even though the size-adjusted dominance of the CBOE bitcoin futures contract ended by March 2018. For checking the arbitrage opportunities in the CME Bitcoin futures, we also use the CME data, which also indicate fewer arbitrage opportunities among the futures and spot markets.

Our study is related to the broad empirical literature on market efficiency in the cryptocurrency market. Empirically, the literature provides mixed results regarding efficiency. To the best of our knowledge, Urquhart (2016) was the first to evaluate the informational efficiency of Bitcoin prices, suggesting that Bitcoin markets were inefficient, although these markets have since matured. Several studies (Nadarajah and Chu, 2017; Bariviera, 2017; Tiwari et al., 2018; Sensoy, 2018; Brauneis and Mestel,

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2018; Kaiser, 2018) provide evidence supporting the efficiency of the Bitcoin market.[2] Conversely, recent works (Cheah et al., 2018; Yonghong et al., 2018) argue that inefficiency still prevails in the Bitcoin market. Moreover, some scholars (Cheah and Fry, 2015; Su et al., 2018) suggest that the Bitcoin price contains a bubble component, while others (Al-Yahyaee et al., 2018; Reynolds et al., 2018; Wei, 2018) provide evidence of market inefficiency in terms of the relationship between cryptocurrencies and other assets. Specifically, Makarov and Schoar (2019a, b) attempt to detect the extent of the arbitrage activity in terms of the region or asset class in the cryptocurrency market. Even though Makarov and Schoar (2019a, b) investigate the arbitrage opportunity between different spot markets, the arbitrage between the spot and futures market has not yet been examined.

Our study is also related to the literature on arbitrage between futures and forward prices. According to a survey by Chow et al. (2000), even when the price difference between futures and forward markets can be significant from a statistical standpoint, the magnitudes will be small and may not be significant from an economic perspective. Other studies include Cornell and Reinganum (1981) on foreign exchange, French (1983) on copper and silver, Park and Chen (1985) on currencies and metal, MacKinlay and Ramaswamy (1988) on stock indices, and Grinblatt and Jegadeesh (1996) on Eurodollar contracts. Our study fills a gap in this literature by demonstrating that the difference between the forward and futures price is insignificant or relatively smaller (amounting to only 0.24% of the Bitcoin spot price).

[2] Kaiser (2018) cannot reject the weak-form efficiency while he observes some seasonality in Cryptocurrencies.

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The remainder of the paper is organized as follows. Section 2 discusses Bitcoin futures and the arbitrage condition. Section 3 explains the empirical method, section 4 reports our results, and section 5 presents the robustness check. Section 6 presents concluding remarks.

2. Bitcoin futures market

Bitcoin is the first and most prevalent cryptocurrency in the world. It was launched in early 2009 just after the publication of Nakamoto's (2008) seminal paper on Bitcoin. By way of several events such as the Mt. Gox bankruptcy (Popper and Abrams, 2014) and the Cyprus shock (Farrell, 2013), its capitalization exceeded 300 billion USD at the end of 2017.[3] Today, the total capitalization of cryptocurrencies is still at a high level at approximately 200 billion USD. The capitalization of Bitcoin represents around two-thirds of the total capitalization of all cryptocurrencies.[4]

On December 10, 2017, CBOE, the largest US options exchange, introduced Bitcoin futures to bring many benefits to traders, including transparency, efficient price discovery, deep liquidity, and centralized clearing.[5] Just after the inception of CBOE's futures, CME, the world largest derivatives exchange, launched Bitcoin futures.

(a) The scheme of CBOE and CME futures

[3] Mt. Gox was the largest cryptocurrency exchange in 2013 and is said to have handled more than 70% of all Bitcoin transactions worldwide (McMillan and Metz, 2013). In 2014, it was announced that customers' Bitcoins were missing and possibly stolen. This exchange ceased all transactions, its website disappeared, and it went into bankruptcy. Although this event was said to lower Bitcoin prices by 23% (Greenberg, 2014), it also attracted public attention to cryptocurrencies and led to more visibility.

[4] As of October 5, 2019

[5] See http://cfe.cboe.com/cfe-products/xbt-cboe-bitcoin-futures

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The basic scheme of CBOE and CME futures is almost identical. Figure 1 shows the price of Bitcoin futures in CBOE and CME on a daily basis, indicating almost identical movements of the price (i.e., the correlation is around 1.0). Both of the futures are cash-settled although the method of settlement is different.

(Figure 1: The price of CBOE and CME's Bitcoin Futures)

The customer's margin requirement is 40% of the daily settlement for CBOE and 35% for CME, reflecting the relatively high volatility of the Bitcoin price. While the trigger conditions are slightly different, both futures markets have circuit breakers. The tick value per Bitcoin is 10 USD for CBOE and 5 USD for CME.

(b) The arbitrage condition using CBOE data

For testing the arbitrage condition between the forward and futures markets in CBOE, the CME data is less than ideal because the reference price of the underlying Bitcoin in CME is not determined at the specific point, which makes it difficult to match the timing of the arbitrage. CME Bitcoin futures are based on the CME Bitcoin Reference Rate (BRR), which aggregates Bitcoin trading activity across major Bitcoin spot exchanges between 3 PM and 4 PM London time.[6]

By contrast, the reference price of the underlying Bitcoin in CBOE is determined at 4 PM ET. The reference price in CBOE is based on Bitcoin price traded in the Gemini Trust Company LLC (Gemini). Gemini is a leading cryptocurrency exchange in New

[6] https://www.cmegroup.com/education/bitcoin/cme-bitcoin-futures-frequently-asked-questions.html

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York that deals with major cryptocurrencies: Bitcoin, Etherium, Zcash, Bitcoin Cash, and Litecoin. The important feature of this exchange is that Bitcoin are priced through the Gemini auction at 4 PM ET, and Bitcoin futures rely on this auction price for settlement. Gemini releases the auction price on a daily basis, allowing us to match the actual timing of the arbitrage by using the 4 PM ET price of Bitcoin futures in CBOE.

(c) Bitcoin futures' trading after June 2019.

Since June 19, 2019, CBOE has stopped adding new futures, so trading of the CBOE Bitcoin futures has ceased. Thus, the availability of the time series is limited as long as we only use the CBOE data. Therefore, we also use the CME data for the robustness check.

Figure 2 (a) shows the trade volume of CBOE and CME futures. The trading activity of CBOE far exceeds that of CME especially at the beginning of the sample and the introduction of the futures. As the figure shows, the trade volume of CBOE is zero after June 19, 2019, when CBOE stopped listing new futures. The contract unit of CBOE futures is 1 BTC, which is smaller than that of CME futures, which is 5 BTC. Figure 2 (b) adjusts for this difference.

(Figure 2: The trade volume of CBOE and CME's Bitcoin Futures)

Table 1 shows the descriptive statistics of Bitcoin futures' returns in CBOE and CME. Ljung-Box statistics reveal that the daily return of Bitcoin futures is not autocorrelated.

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(Table 1: The fundamental statistics of the return of Bitcoin futures in CBOE)

3. Empirical method

3.1 The arbitrage condition between forward and futures

Our empirical method follows the previous literature on testing for the efficiency in futures market.

To obtain the Bitcoin forward price, we use the cost of the carry relationship as follows:

$$G_t = S_t \exp\,(r_t T), \tag{1}$$

where G_t is the price of the forward contract at time t, S_t is the price of the underlying asset at time t, and $\exp(r_t T)$ is the risk-free rate of return over the period from t to T.

We define the deviation from forward and future as:

$$Deviation_t = |G_t - H_t|, \tag{2}$$

where H_t is the price of the futures contract at time t. The bid-ask spread is considered as the cost of arbitrage, that is, so there is no arbitrage opportunity as long as this deviation is smaller than the bid-ask spread. For testing the no arbitrage condition, we investigate whether the deviation from the forward and futures is smaller than the bid-ask spread of the forward and futures market.

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$$Hypothesis: no\ arbitrage\ when\ Deviation_t\ is\ smaller\ than$$
$$the\ bid - ask\ spread$$

3.2 The arbitrage condition in Bitcoin market

To test the hypothesis proposed in section 3.1, we consider arbitrage pertaining to: (i) selling (buying) Bitcoin futures at 4 PM ET, and (ii) buying (selling) Bitcoin forward traded in Gemini at 4 PM ET. Figure 3 indicates the time flow of this arbitrage when we use the intraday data of Bitcoin futures in CBOE. For example, investors short the Bitcoin futures at time *t* (4 PM ET). In this case, the timing of the cash settlement of the futures is time *T* (4 PM ET), which is the maturity of Bitcoin futures in CBOE. However, the investor also enters into a forward contract of Gemini at time *t* (4 PM ET) with maturity *T*, that is, the same length as that of the futures in CBOE. At time *T*, the investors deliver at the price of Gemini at 4 PM ET because they would receive Gemini at 4 PM ET from the futures contract.

(Figure 3: Arbitrage condition between Bitcoin futures and forwards)

3.2 The deviation under no arbitrage condition

The daily marking to market of gains and losses on open futures positions causes futures prices to diverge from forward prices as the daily gains or losses can be invested or borrowed. Cox et al. (1981) show that if the price of the underlying asset and that of the risk-free discount bond are positively (negatively) correlated, then the future prices should be larger (smaller) than the forward prices. However, with small covariances

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between the Bitcoin price and the risk-free rate, the authors predict that the price of futures and forwards should be indistinguishable.

Following Cornell and Reinganum (1981), we estimate the covariance between the price change of the Bitcoin futures price and the yield difference of the risk-free rate, which is about 9×10^{-7} (correlation is 0.05). In addition to the covariance estimations, we regress the price change of the futures price on the yield difference of the risk-free rate:

$$\Delta price_t = \alpha + \beta \cdot \Delta\, riskfree_t + \varepsilon_t, \tag{3}$$

Table 2 shows the estimation results of eq. (3), indicating no significant relationship (t-statistics is 1.27).[7] Therefore, the futures and forward prices should be nearly identical in the Bitcoin market.

(Table 2: The estimation result)

3.3 Determinants of the discrepancy

To detect the determinants of the discrepancy between futures and forward prices, we estimate the following regression:

$$Deviation_t = \alpha + \beta \cdot\ Bidaskspread_t + \gamma \cdot turnover_t + \varepsilon_t, \tag{3}$$

[7] The standard error is computed based on Newey and West (1987).

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where $Deviation_t$ is the absolute value of the difference in futures and forward prices, $Bidaskspread_t$ is the bid-ask spread in the price of Bitcoin futures, $turnover_t$ is the turnover in Bitcoin futures, and ε_t is the error term. For $turnover_t$ we use the turnover of Bitcoin futures in the period from 4 PM to 4:05 PM ET, which covers the auction time in Gemini. We use a different measure for checking the robustness.

Our prediction in equation (3) is that $Bidaskspread_t$ has a positive effect on the deviation while $turnover_t$ has a negative effect. When $Bidaskspread_t$ increases, the cost of the arbitrage should increase for investors, suggesting the $Bidaskspread_t$ has a positive effect on the deviation between the forward and futures price. However, when $turnover_t$ increases, this improves liquidity and the deviation should therefore decrease.

We choose these explanatory variables based on previous research. First, since Wei (2018) reveals that the inefficiency of the Bitcoin market is related to the liquidity measure, it is natural to include the bid-ask spread and turnover to explain the movement of the deviation. Second, Makarov and Schoar (2019a, b) show that order flow plays an important role in explaining the size of the arbitrage spreads between cryptocurrency exchanges; therefore, we also include market turnover to explain the deviation.

4. Empirical result

4.1 Data

We use the data of CBOE futures price on active contracts and Bitcoin spot prices traded in Gemini from Bloomberg. Bloomberg provides intraday data of the Bitcoin

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futures price hence, we use the 4 PM ET price of futures to match the timing of the Gemini price.

We obtain the bid and ask price of Bitcoin futures at 4 PM ET from Bloomberg, which is the actual trading cost of the investor engaging in the arbitrage. The Gemini price is determined in the auction hence, the bid-ask price of Gemini does not exist. We also obtain the trade volume of Bitcoin futures with corresponding time step from Bloomberg. The trade volume of Bitcoin traded in Gemini is obtained from Gemini's website.

Our dataset covers the period December 2017 to December 2018. The number of observations included in our analysis is relatively larger than previous studies based on intraday data, for example, Dyhrberg et al. (2018).

4.2 Empirical result of the arbitrage condition

Figure 4 shows the time series of the prices of Bitcoin futures, forwards, and differentials. This figure indicates that the deviation between the futures and forward prices was wide (1) during December 2017 to February 2018 and (2) during November 2018 to December 2018, both of which are commonly referred to as cryptocurrency crashes (Williams-Grut, 2018; Chambers, 2018). Figure 5 depicts the time series of 1-month returns, which shows a significant drop during these periods. We define "crash period" by more than 40% drop within one trading month.

(Figure 4: Future and forward prices compared to the bid-ask spread of futures)

(Figure 5: Time series of Bitcoin return (1-month return))

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This observation can be confirmed via a simple estimation. Table 3 (a) shows the result of the test for equality of averages between the deviation between forward and futures price and the bid-ask spread in the Bitcoin market in CBOE. The left part of Table 3 (a) indicates that the mean discrepancy between futures and forward prices is significantly different from the bid-ask spread using the whole sample. Table 3 (b) shows the average deviation and bid-ask spread for each period and indicates that the deviation is larger than the bid-ask spread, which suggests that there is an arbitrage opportunity during this period.

However, we find few arbitrage opportunities during the normal time (the period without crisis). Table 3 (a) shows the deviation is not significantly different from the bid-ask spread during March 2018 to October 2018, which coincides with the normal time. Therefore, we can interpret this as the arbitrage being sufficient during the normal period.

Conversely, Table 3 (a) shows that the mean deviation between futures and forward prices is significantly different from the bid-ask spread when we use the data during (a) December 2017 to February 2018 and (b) November 2018 to December 2018, suggesting that arbitrage is limited to the crash time.

(Table 3: Test for equality of medians between the deviation between forward and futures and the bid-ask spread in Bitcoin markets)

Even though there seems to be an arbitrage opportunity during the entire period, the economic significance of the deviation between futures and forwards is not large. Table 3 (b) reports the average deviation and bid-ask spread for each period when the average

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deviation is 16.69 USD and the average bid-ask spread is 36.10 USD (i.e., the difference is 19.41 USD). During this period, the average price of Bitcoin futures is 8,003 USD, therefore the difference only amounts to 0.24%. In addition, the deviation of futures and forwards is statistically not different from the bid-ask spread with respect to the median. Table 4 reports the result of the test for equality of medians between the deviation among forwards and futures and the bid-ask spread in the Bitcoin market. This table indicates the median of the deviation is not significantly different from that of the bid-ask spread, except for November 2018 to December 2018 when the median of deviation is smaller than that of the bid-ask spread.

(Table 4: Test for equality of medians between the deviation between forwards and futures and the bid-ask spread in Bitcoin market)

4.3 Empirical results of the determinants of the deviation

Our regression results based on eq. (3) are reported in Table 5 (a), demonstrating that the bid-ask spread of Bitcoin futures has a positively significant relationship with the deviation of futures and forwards prices at the 5% significance level. This result implies that the deviation widens when the bid-ask spread increases, consistent with our prediction. This result is robust when we use the data from March 2018 to October 2018, as shown in Table 5 (b).

(Table 5: Estimation results)

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In Table 5, we also report that the turnover does not have a significant relationship with the deviation. The correlation between the bid-ask spread and the turnover of Bitcoin futures is only -0.048; thus, we can rule out the possibility of multicollinearity. This result is different from Makarov and Schoar (2019a, b).

To check whether a different measure of turnover has a significant relationship, we include the turnover of Bitcoins traded in Gemini. The result is reported in Table 6 and suggests that the turnover does not have a significant relationship with the deviation (the correlation between the turnover of Bitcoin futures and Bitcoin traded in Gemini is only 0.012).

(Table 6: Estimation results)

5. Robustness check: CME data

As we describe in section 2, the main problem with using CBOE market data is short life span. Therefore, it is important to use CME data for testing the arbitrage condition by using the relatively longer data.

5.1 Data

We obtain the data, which is daily, of CME futures price on active contracts and BRR from Bloomberg. We compute the forward price of Bitcoin based on eq. (2) described in section 4. We also again use the bid and ask price of Bitcoin futures obtained from Bloomberg.[8]

[8] We exclude data from three days (2018/4/23, 2018/4/23, and 2018/5/1) as the ask prices from Bloomberg are extremely large (the ask price is more than twice as high as the bid price).

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5.2 Empirical result of the arbitrage condition

First, we check whether the deviation between the forwards and futures in the CME market is different from the bid-ask spread of Bitcoin futures. Figure 6 shows the time series of the deviation between the forward and futures with the bid-ask spread in the CME market. This figure indicates that the fluctuation of the deviation and bid-ask spread of CME is more volatile than that of CBOE. The correlation between these variables is 0.17. The average of deviation is 634 USD while the average of the bid-ask spread is 709 USD, thus the deviations are within the bid-ask spread.

(Figure 6: Time series of the deviation from forward and futures of Bitcoin with the bid-ask spread)

To observe the relationship between the deviation and the bid-ask spread in CME, Table 7 reports the results of whether the mean deviation of futures and forwards prices is statistically different from the bid-ask spread. This table indicates that the deviation is significantly different from the bid-ask spread and the average of deviation is smaller than the bid-ask spread. This result implies few arbitrage opportunities among forwards and futures in the Bitcoin market when we use CME data.

(Table 7: Test for equality of mean between the deviation between forwards and futures and the bid-ask spread in the Bitcoin market in the case of CME)

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To examine how the deviation is affected by the liquidity measure, we also run the regression based on eq. (3). As we use the data of CBOE, $Deviation_t$ is the deviation between forward and futures and $Bidaskspread_t$ is the bid-ask spread in the price of Bitcoin futures. We use the daily turnover of Bitcoin futures in CME for $turnover_t$.

Table 8 shows that our empirical result is consistent with our prediction and the result obtained from the CBOE data. When we regress the deviation on the bid-ask spread, we find that the coefficient is positive and significant at the 10% level, which is consistent with the prediction. Regressing the deviation on the turnover, we find that the coefficient is negative and significant at the 1% level, which is also consistent with our prediction. Finally, we regress the deviation on the bid-ask spread and turnover. The coefficient of the bid-ask spread is positively significant at the 10% level while that of the turnover is negatively significant at the 5% level.

(Table 8: Estimation results)

6. Conclusion

We empirically test whether investors arbitrage between the Bitcoin spot and futures market. There is a significant body of literature considering the arbitrage between futures and their underlying assets. This motivates us to evaluate arbitrage opportunities in the Bitcoin market after the launch of Bitcoin futures. Our study shows few arbitrage opportunities especially in the normal period, while market crashes lead to arbitrage opportunities.

From our empirical result, we interpret the relationship between the market crash and the introduction of Bitcoin futures at the end of 2017. It is important to evaluate the

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arbitrage opportunities using the spot and futures price to understand the Bitcoin's price movement during the market crash at the end of 2017. Some studies argue that the introduction of the futures market led to a sharp decline in the Bitcoin price (see Hale et al., 2018). However, there must be a strong link between the two markets for the spot market to crash via the futures market. If the link is weak, pressure from the futures market would not affect the Bitcoin spot market. Therefore, empirical research on the arbitrage between these two assets should provide evidence if the launch of the new financial instrument did, indeed, trigger the market crash. Our empirical study finds no linkage between these markets during the crash period, casting doubt over the commonly held belief that the introduction of the Bitcoin futures market has negatively affected the Bitcoin spot market.

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Table 1 The fundamental statistics of the return of Bitcoin futures in CBOE

	Mean	Max	Min	Std. Dev.	Skewness	Kurtosis	LQ	Obs
CBOE	-0.0006	0.2415	-0.1989	0.0478	0.2044	6.3548	0.658	397
CME	-0.0006	0.2492	-0.2182	0.0496	0.1636	7.0026	0.125	466

Note: This table provides the fundamental statistics of the return of Bitcoin futures. The frequency of the data is daily. LB is the p-value of the Ljung-Box statistics. Source: Bloomberg.

Table 2 Estimation results

	price change in the futures price
α	-0.01
	(0.003)
β	8.326
	(6.569)
Num of Obs	263

Notes: This table presents the estimation results. The dependent variable is the price change in the futures price. The independent variable is constant and the yield difference of the risk-free rate. The OLS regression coefficients are also reported. The robust standard error with Newey-West is in parentheses (*** p<0.01, ** p<0.05). The data were produced daily from December 2017 to December 2018.

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Table 3 Test for equality of averages between the deviation between forward and futures and the bid-ask spread in Bitcoin market in the case of CBOE

(a) Test statistic

	Whole sample	Sub sample		
	Dec, 2017 – Dec, 2018	Dec, 2017 – Feb, 2018	March, 2018 – Oct, 2018	Nov, 2018 – Dec, 2018
t-statistics	3.106	2.609	1.761	3.281
(p-value)	(0.002)	(0.010)	(0.079)	(0.002)

(b) The average value of the deviation and bid-ask spread

	Whole sample	Sub sample		
	Dec, 2017 – Dec, 2018	Dec, 2017 – Feb, 2018	March, 2018 – Oct, 2018	Nov, 2018 – Dec, 2018
Deviation	16.69	18.55	12.53	11.58
Bid-ask spread	36.10	204.65	14.98	31.67

Note: This table provides information about the test for equality of averages between the deviation between forwards and futures and the bid-ask spread in Bitcoin market. The CBOE data is used. This table shows the result of whether the deviation of futures and forward prices is different from that of the bid-ask spread. In (a) test statistics, P-values are in parentheses. The timing of futures and forward prices is 4 PM ET. The data frequency is daily.

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Table 4 Test for equality of medians between the deviation between forward and futures and the bid-ask spread in Bitcoin market in the case of CBOE

(a) Test statistic

	Whole sample	Sub sample		
	Dec, 2017 – Dec, 2018	Dec, 2017 – Feb, 2018	March, 2018 – Oct, 2018	Nov, 2018 – Dec, 2018
t-statistics	1.121	0.777	1.761	3.281
(p-value)	(0.262)	(0.437)	(0.079)	(0.002)

(b) The median value of the deviation and bid-ask spread

	Whole sample	Sub sample		
	Dec, 2017 – Dec, 2018	Dec, 2017 – Feb, 2018	March, 2018 – Oct, 2018	Nov, 2018 – Dec, 2018
Deviation	10.00	30.00	10.00	10.00
Bid-ask spread	14.83	36.14	11.03	16.23

Note: This table provides information about the test for median of averages between the deviation between forward and futures and the bid-ask spread in the Bitcoin market. The CBOE data is used. This table shows the result of whether the deviation of futures and forward prices is different from that of the bid-ask spread. In (a) test statistics, P-values are in parentheses. The timing of futures and forward prices is 4 PM ET. The data frequency is daily.

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Table 5 Estimation results

(a) Whole sample (Dec 2017–Dec 2018)

	(1)		(2)	(3)	
Bid-ask spread	3.623	**		3.616	**
	(1.774)			(1.767)	
Turnover			-0.235	-0.160	
			(0.279)	(0.209)	
R-squared	0.268		-0.001	0.266	
No. of observations	265		265	265	

(b) Subsample (Mar 2018–Oct 2018)

	(1)		(2)	(3)	
Bid-ask spread	0.287	**		0.288	**
	(0.118)			(0.120)	
Turnover			-0.021	-0.023	
			(0.036)	(0.035)	
R-squared	0.023		-0.005	0.018	
No. Of observations	172		172	172	

Notes: This table shows the estimation results using the CBOE data. The dependent variable is the deviation of futures and forward prices. The independent variable is constant, bid-ask spread of Bitcoin futures and turnover. The OLS regression coefficients are also reported. The robust standard error with Newey-West is in parentheses (*** p<0.01, ** p<0.05, * p<0.10). The data were produced daily from December 2017 to December 2018.

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Table 6 Estimation results

(a) Whole sample (Dec 2017–Dec 2018)

	(1)	(2)	(3)
Turnover (Bitcoin Futures)	-0.235		-0.226
	(0.279)		(0.272)
Turnover (Gemini)		-0.050	-0.048
		(0.049)	(0.047)
R-squared	-0.001	0.000	-0.002
No. of observations	265	260	260

(b) Subsample (Mar 2018–Oct 2018)

	(1)	(2)	(3)
Turnover (Bitcoin futures)	-0.035		-0.040
	(0.038)		(0.040)
Turnover (Gemini)		0.010	0.011
		(0.012)	(0.012)
R-squared	-0.004	-0.002	-0.005
No. of observations	150	145	145

Notes: This table shows the estimation results using the CBOE data. The dependent variable is the deviation of futures and forward prices. The independent variable is constant, the turnover of Bitcoin futures and that of Bitcoin traded in Gemini. The OLS regression coefficients are also reported. The robust standard error with Newey-West is in parentheses (*** p<0.01, ** p<0.05, * p<0.10). The data were produced daily from December 2017 to December 2018.

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Table 7 Test for equality of mean between the deviation between forward and futures and the bid-ask spread in Bitcoin market in the case of CME

(a) Test statistic

t-statistics	2.099
(p-value)	0.36

(b) The average value of the deviation and bid-ask spread

Deviation	634
Bid-ask spread	709

Note: This table provides information on the price differential between futures and forward prices in the Bitcoin market and the bid-ask spread of the Bitcoin futures market. The CBOE data is used. This table reports the result of whether the deviation of futures and forward prices is different from that of the bid-ask spread. P-values are in parentheses. The timing of futures and forward prices is 4 PM ET. The data were produced daily from February 2018 to August 2019.

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Table 8 Estimation results

	(1)		(2)		(3)	
Bid-ask spread	0.142	*			0.125	*
	(1.947)				(1.709)	
Turnover			-0.026	***	-0.020	**
			(2.661)		(2.267)	
R-squared	0.027		0.015		0.034	
Num of Obs	403		403		403	

Notes: This table shows the estimation results using the CME data. The dependent variable is the deviation of futures and forward prices. The independent variable is constant, bid-ask spread of Bitcoin futures and turnover. The OLS regression coefficients are also reported. The robust standard error with Newey-West is in parentheses (*** p<0.01, ** p<0.05, * p<0.10). The data were produced daily from February 2018 to August 2019.

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Figure 1 The price of CBOE and CME's Bitcoin futures

Source: Bloomberg

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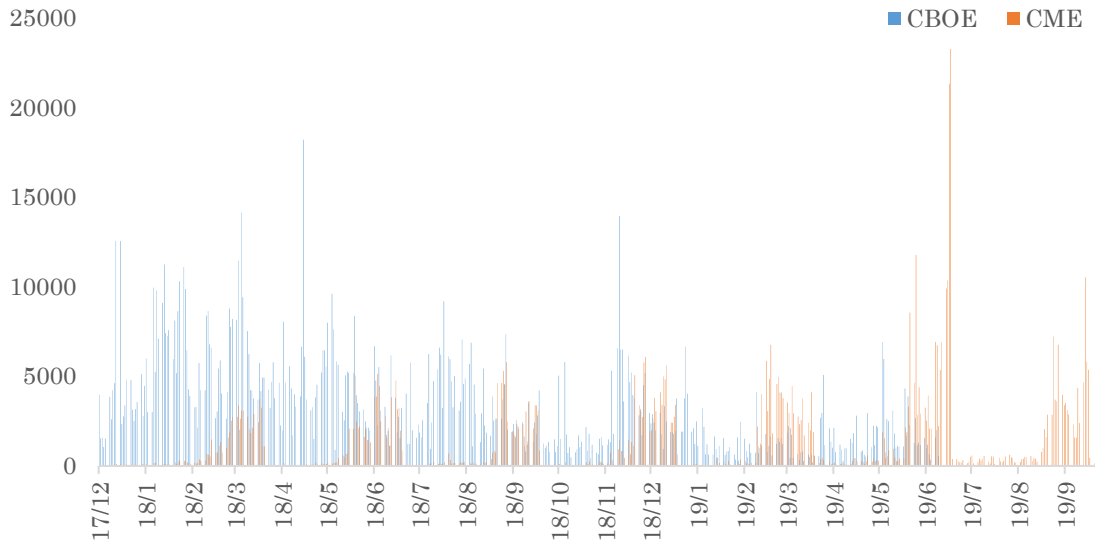
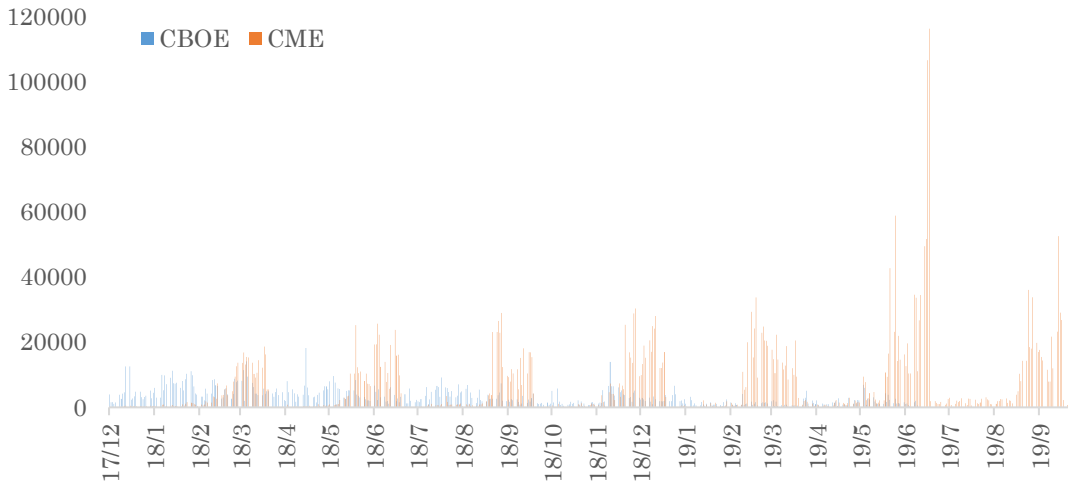
Figure 2 The trade volume of CBOE and CME's Bitcoin Futures

(a) Trading volume without adjustment



(b) Trading volume with adjustment



Note: In Figure 2 (ii), the contract unit of Bitcoin futures in CME and CBOT is adjusted. The contract unit of CBOE futures is 1 BTC while the unit of CME futures is 5 BTC.

Source: Bloomberg

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Figure 3 Arbitrage condition between Bitcoin futures and forwards

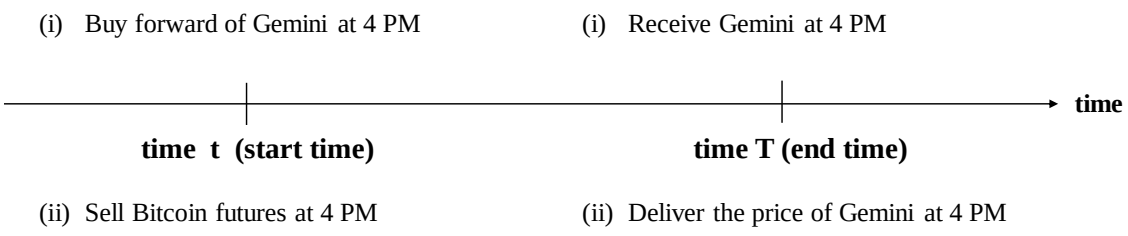
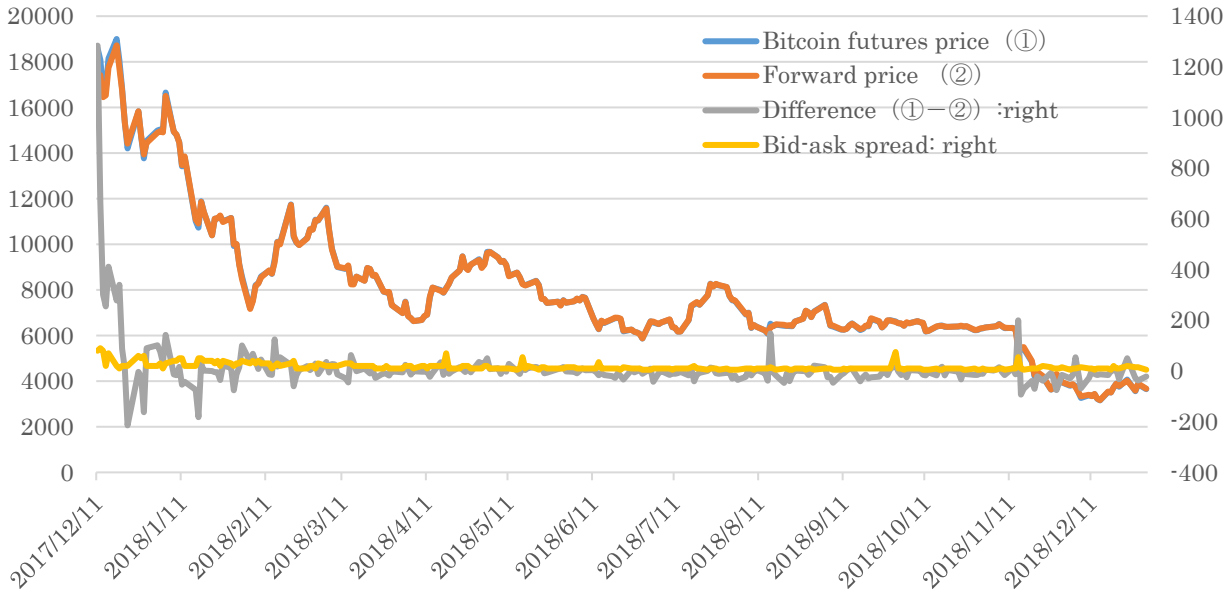


Figure 4 Futures and forward prices compared to the bid-ask spread of futures



Note: This figure describes the Bitcoin futures, forward, differential, and bid-ask prices. The sample period is from December 11, 2017 to December 28, 2018. The timing of futures and forward prices is 4 PM ET and the data frequency is daily.

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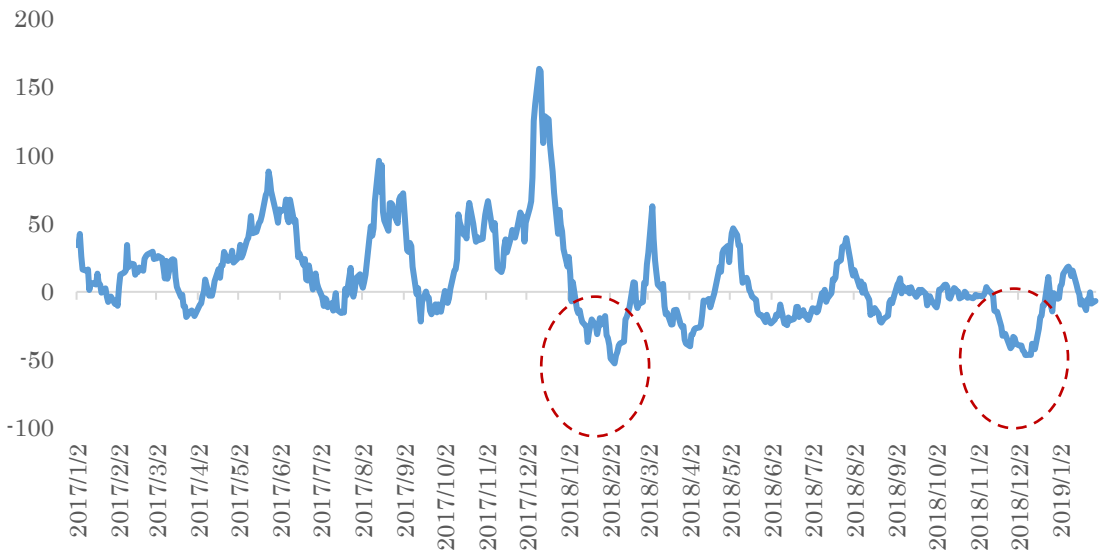
Figure 5 Time series of Bitcoin return (1-month return)



Source: Bloomberg

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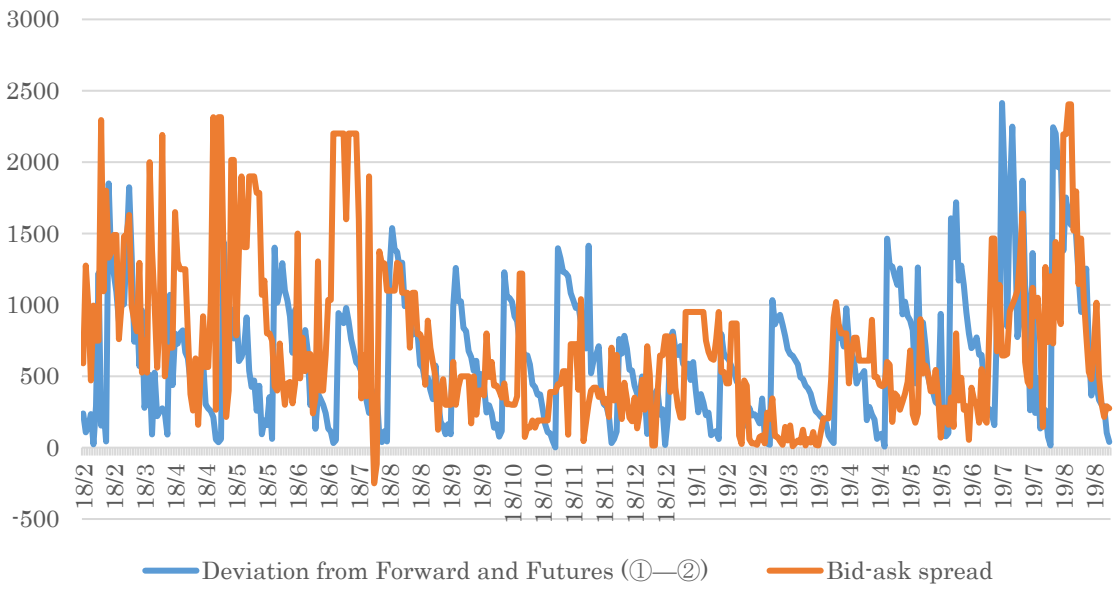
Figure 6 Time series of the deviation from forward and futures of Bitcoin with the bid-ask spread



Note: The deviation of the forward and future price and the bid-ask prices. The CME data is used. The sample period is from February 9, 2018 to August 30, 2019 and the data frequency is daily.

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